

Sirisha P R

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Experienced Data Analyst with 8 years in data analytics, statistical modeling, ETL/ELT, and BI tools across industries. Proficient in SQL, Python, R, Scala, PySpark, and advanced analytics tools, with expertise in cloud platforms (AWS, Azure, GCP) and relational/NoSQL databases. Skilled in building scalable pipelines, delivering actionable business insights, and transforming complex data into impactful solutions.

TECHNICAL SKILLS

Programming Languages & Big Data Frameworks: SQL, R, Python, Scala, T-SQL, NoSQL, PySpark, Apache Spark, TensorFlow, PyTorch, Hadoop, Spark, MapReduce

Database: Oracle, MySQL, Snowflake, PostgreSQL, MongoDB, SQL Server

Statistical Techniques: Confidence Intervals, Hypothesis Testing, A/B Testing, Statistical Tests, ANOVA, Probability distribution, statistical methods, forecasting, predictive analytics, data mining, quantitative analysis, descriptive analysis, statistics

Machine Learning & AI: Regression (Linear, Logistic), Classification, Regularization (Lasso, Ridge), Ensemble Methods (Bagging, Boosting), Clustering (K-Means, Hierarchical), Time Series Analysis, Natural Language Processing (NLP), Deep Learning

Cloud Platforms: AWS (EC2, S3, Redshift, Glue, Athena, Aurora, EMR, Kinesis, Lambda, DynamoDB, QuickSight, Data Firehose), Microsoft Azure (Data Factory, Data Explorer, Synapse, Data Lake, Functions), Google Cloud Platform (GCP) (BigQuery, Dataflow, Dataproc, Vertex AI, Looker)

Data Science & Analytics Packages: Scikit-Learn, dplyr, caret, ggplot2, Pandas, Numpy, Matplotlib, Seaborn, Plotly, ARIMA, Polars, H2O, SciPy

Business Intelligence (BI) & data visualization Tools: Tableau, Power BI, SAS Visual Analytics, Looker, QuickSight

IDEs: Visual Studio Code, PyCharm, Jupyter Notebook, IntelliJ

Data Engineering: ETL pipelines, ELT, data transformation, data modeling, data warehouse, performance optimization, BI solutions, root cause analysis (RCA), data ingestion, data integration, data interpretation, data pipeline, data governance, Alteryx

Orchestration, DevOps & Automation Tools: Databricks, Apache Airflow, dbt, Docker, Kubernetes, CI/CD Pipelines, Git, GitHub Actions, version control, Power Apps, Power Automate, Automation (Macros, VBA, Power Query)

Project Management & Other Tools: JIRA, cross-functional collaboration, Google Analytics, Adobe analytics, Salesforce, CRM, ERP, Advanced Excel (Pivot Tables, VLOOKUP, XLOOKUP, Index & Match, SUMIFS, Data Validation), documentation, key performance indicators monitoring & reporting, Cohort analysis, ServiceNow, SaaS, SSIS, SSRS, SDLC, Agile, Waterfall

Business & Interpersonal skills: process improvement, problem-solving, business recommendations, attention to detail, analytical thinking, strategic decision making, detail-oriented, stakeholder collaboration, communication skills, storytelling with data

Certifications: [Tableau Desktop Specialist Certification](#)

PROFESSIONAL EXPERIENCE

Data Analyst – Baylor Scott & White Health, TX

Aug 2024 – Present

- Directed the development of an AI-powered claims monitoring and anomaly detection system for Baylor Scott & White Health, leveraging AWS Lambda for real-time analysis of medical claims. This solution enhanced early detection of fraud, waste, and abuse (FWA) across the provider network, supporting value-based care initiatives and improving payer-provider transparency.
- Conducted **Exploratory Data Analysis (EDA)** in Python to uncover patterns and correlations between claim anomalies, provider billing behaviors, claim types, and temporal trends - informing feature engineering and enhancing fraud detection model accuracy.
- Designed and deployed interactive **Power BI dashboards** to track key operational and clinical metrics, including claims denial rates, fraud risk scores, patient risk stratification, and provider billing patterns, empowering compliance and investigation teams with real-time, actionable insights.
- Built distributed **ETL pipelines** in PySpark to process large volumes of claims and EHR data (millions of records) from S3 into a unified, analytics-ready format.
- Applied **BERT** data models to analyze textual fields such as diagnosis descriptions, procedure justifications, and appeal notes, enabling semantic understanding beyond keyword matching.

- Used **Random Forest classifier** to predict fraudulent healthcare claims by learning complex, non-linear patterns across features like claim amount, diagnosis-procedure consistency, and provider history.
- Leveraged SecureGPT to accelerate the generation of Python-based analytics pipelines while enforcing data access controls, reducing model development and data processing time by 40%. All workflows were aligned with Baylor Scott & White Health's security and compliance framework.
- Deployed **XGBoost** for high-performance fraud classification, leveraging its gradient boosting algorithm to detect subtle anomalies in claims data.

Program Manager (Pricing Analytics) – Amazon, India

Dec 2021 – Aug 2023

- Facilitated **Agile ceremonies** such as daily stand-ups and sprint planning, ensuring effective collaboration and project progress
- Developed **business intelligence reports** using **QuickSight** for demand, competitor price matching, Pareto analysis, and inventory level reports weekly, saving approximately 30 man-hours per week
- Collaborate and developed the project to **optimize Ship-cost through Unit Per Box (UPB) metrics** like box utilization ratio and air percentage, reducing waste by 15%. Delivered insights and reports via QuickSight, S3, and Athena
- Conducted end-to-end cost analysis by **optimizing complex SQL queries** to process 1.5M rows of data. Built interactive **cost-to-serve dashboards** and achieved a 12% reduction in cost estimation errors
- Automated **VBA macro** in Excel using **Power Query** & performed analysis on QuickSight to calculate team productivity, which improved business performance by enhancing resource management & reducing ~6 hr/week of manual work

Data Analyst – Amazon, India

Aug 2016 – Nov 2021

- Engineered 20+ scalable **ETL pipelines** and supported data warehousing solutions using **AWS Glue, Redshift, and S3**, boosting processing efficiency by 30%. Improved **data modeling** and transformation pipelines using AWS Glue Studio
- Conducted **funnel analysis** to identify checkout bottlenecks by tracking drop-off points. **Reduced cart abandonment** by 20% in three months, boosting checkout completions and monthly revenue
- Designed **Item Found Glance View metrics** for competitor-matched products and built dashboards for overlap analysis. Delivered insights that enhanced pricing by 10%, **boosting customer retention**
- Visualized dashboards to assess website traffic metrics/KPIs of competitors across the EU region, **increasing user retention by 20%** and refining campaign workflows
- Developed complex data wrangling scripts using **Python libraries** such as **pandas, numpy** and **datetime** to clean, reshape, and enrich large datasets, ensuring high-quality inputs for downstream analysis and visualization
- Evaluated and optimized data sources for QuickSight dashboards by configuring **scheduled refreshes**, implementing data blending, calculated fields, and custom aggregations to improve reporting accuracy and efficiency
- Built and maintained **custom dashboards** using **Excel, Python & Matplotlib** to visualize price trends, errors, and SKU-level metrics for business stakeholders, enhancing data driven decision-making
- Leveraged **data validation** frameworks to detect anomalies, outliers, and schema inconsistencies in pricing data, ensuring high **data integrity** across pricing platforms

EDUCATION

Masters in Business Analytics, University of North Texas, USA

Aug 2023 – May 2025

Recipient of Mary D. Walsh Endowed Scholarship

GPA:4/4

Bachelors in Computer Science, Visvesvaraya Technological University, India

Jun 2012 – May 2016